Shayan Shakeri

(416) 471-6651 | shayansnezhad@gmail.com | github.com/sshakerinezhad | merlyn-labs.com | sshakerinezhad.github.io |

Toronto, Canada

PROFESSIONAL SUMMARY

AI Research Engineer focused on post-training, evaluation, and sim-to-real deployment of deep-learning models for robotic manipulation. Conservative VLA finetuning doubled spatial-generalization success (21% to 42%), an open-sourced flow-matching VLA integration lets RLinf run large-scale RL training on the BEHAVIOR-1K suite in OmniGibson, and models run on real hardware including a hand-built AlohaMini embodiment. Co-founder of Merlyn Labs; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

VLA Post-Training | Finetuning | Sim-to-Real Deployment | Reinforcement Learning | Evaluation Systems | PyTorch / JAX

WORK EXPERIENCE

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- **Refine and evaluate deep-learning models for robotic manipulation via VLA post-training and finetuning**; a conservative finetuning recipe doubled position-swap success (21% to 42%), proving spatial generalization over memorization.
- Open-sourced a flow-matching VLA integration for RLinf, enabling large-scale RL training on the BEHAVIOR-1K suite in OmniGibson.
- Building continuous-improvement frameworks: VLM judges that score rollouts into dense, context-dependent RL rewards that resist gaming.
- Running sim-to-real deployment of household manipulation tasks on a hand-built AlohaMini embodiment.

BMO AI Centre of Excellence
Sep 2024 - Present

AI Research Engineer | Toronto, Canada

- **Uncovered systematic bias to downplay investment risk** in a GenAI tool serving $200B+ AUM wealth management.
- Built a deterministic agent eval pipeline using hundreds of synthesized inputs to track model progress and detect misaligned outputs at scale.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop PeriPulse (FDA Breakthrough-designated neurostimulation device, now in multi-national trials); designed and assembled PCBs and authored IEC 60601-1 / ISO 13485 validation protocols.

PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place, Standard Track

Trained manipulation policies on 10,000+ demonstrations across 22 tasks. Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Published a technical report and analysis on LessWrong.

VLA, Reinforcement Learning, Large-Scale Training, OmniGibson, PyTorch

Voice-Controlled Robotic Prosthetic

Developed an LLM-based control pipeline converting natural language into manipulation sequences for a 7-DOF simulated arm. Integrated YOLO object detection with LiDAR depth mapping for 3D localization and grasp planning.

EDUCATION

M.Eng in AI & Robotics, University of Toronto — Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Robot Foundation Models & Deployment: VLA Models, Post-Training, Finetuning, Behavior Cloning, Reinforcement Learning, Imitation Learning, Sim-to-Real, ROS2, Model Deployment

Evaluation, Simulation & Languages: Deterministic Eval Pipelines, VLM Judges, Continuous-Improvement Frameworks, OmniGibson, MuJoCo, Flow Matching, Python, C/C++, PyTorch, JAX